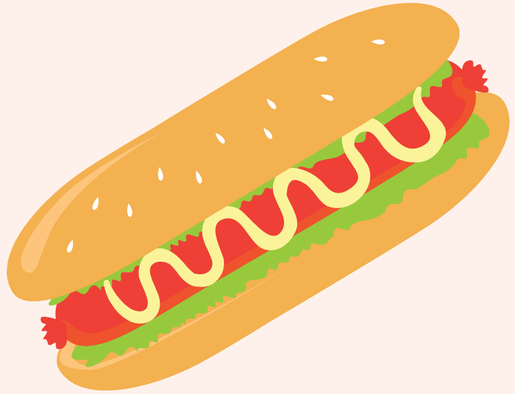

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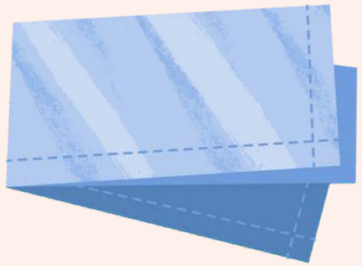
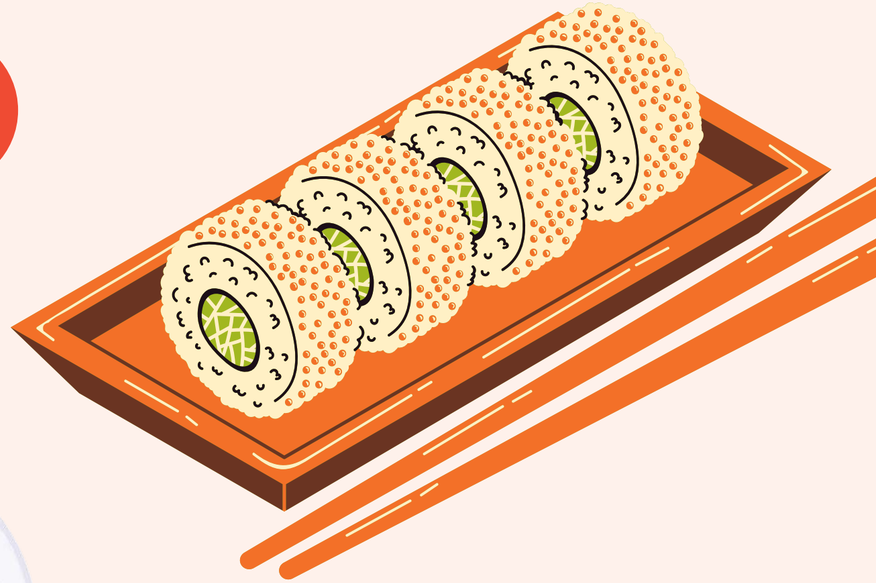


FOUNDATIONS OF DEEP LEARNING

Food image classification

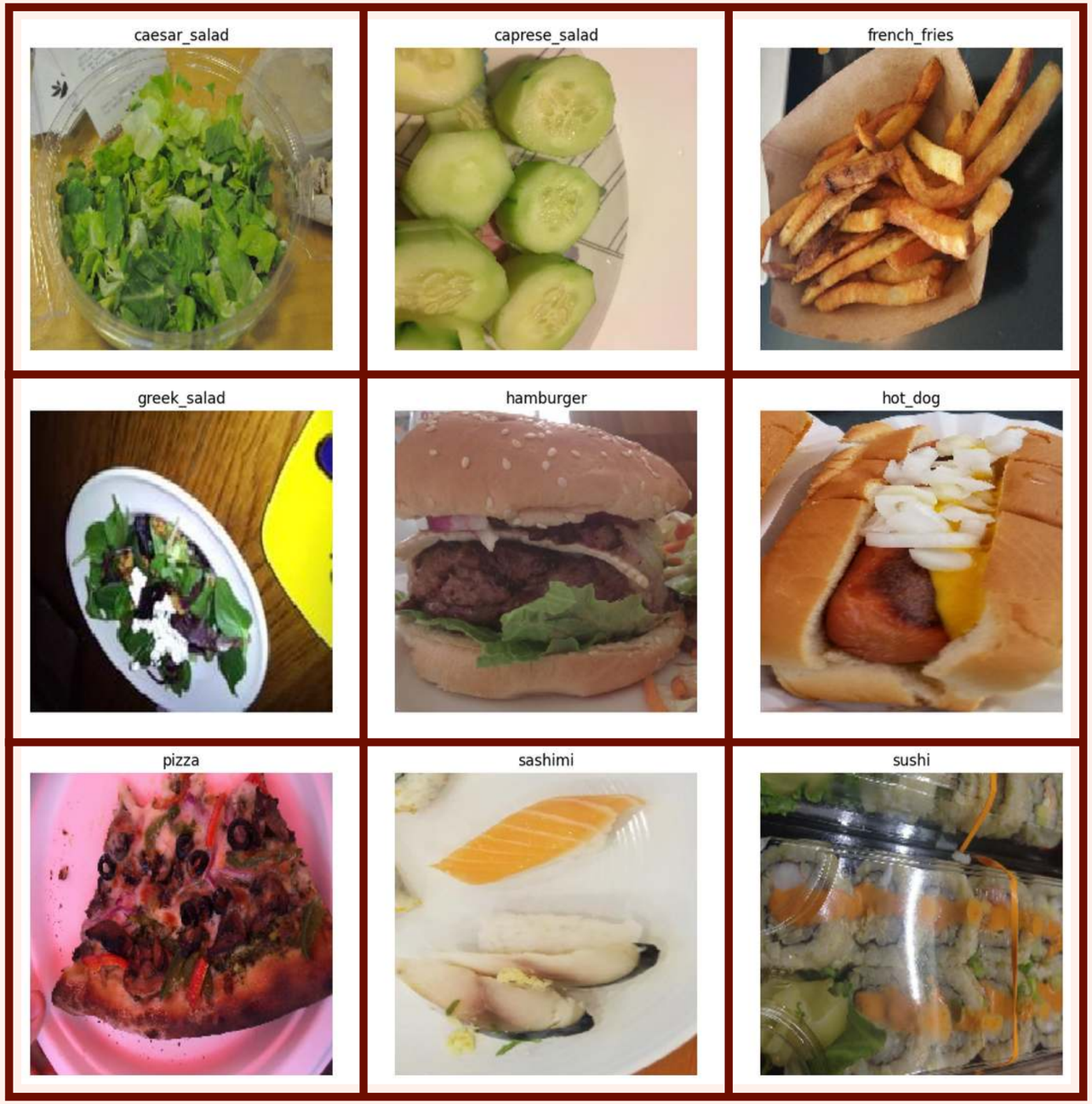
JAMIL AMAIZA 885861



INTRODUCTION

The project focuses on image classification within a diverse food dataset containing 978 images across 9 distinct categories:

- Sushi (124 images)
- Pizza (299 images)
- Sashimi (40 images)
- Hot Dog (31 images)
- Hamburger (238 images)
- French Fries (181 images)
- Greek Salad (24 images)
- Caesar Salad (26 images)
- Caprese Salad (15 images)



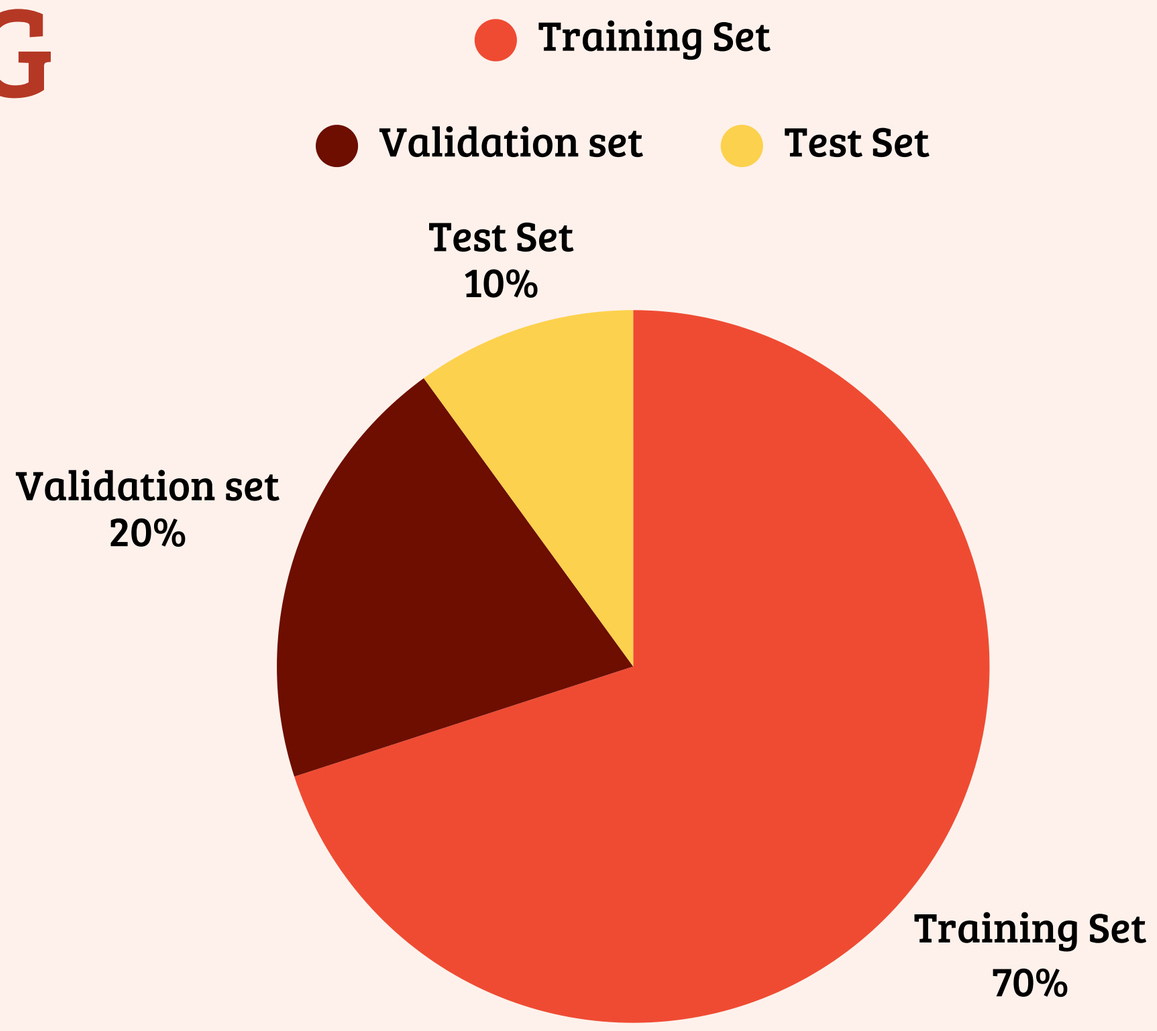
GOAL OF THE PROJECT

The primary goal is to accurately classify each food image into its corresponding category. To achieve this, we will develop a Convolutional Neural Network (CNN) model from scratch and explore various techniques to enhance its performance. Finally, we will benchmark our custom model against a pretrained CNN to compare accuracy and effectiveness.

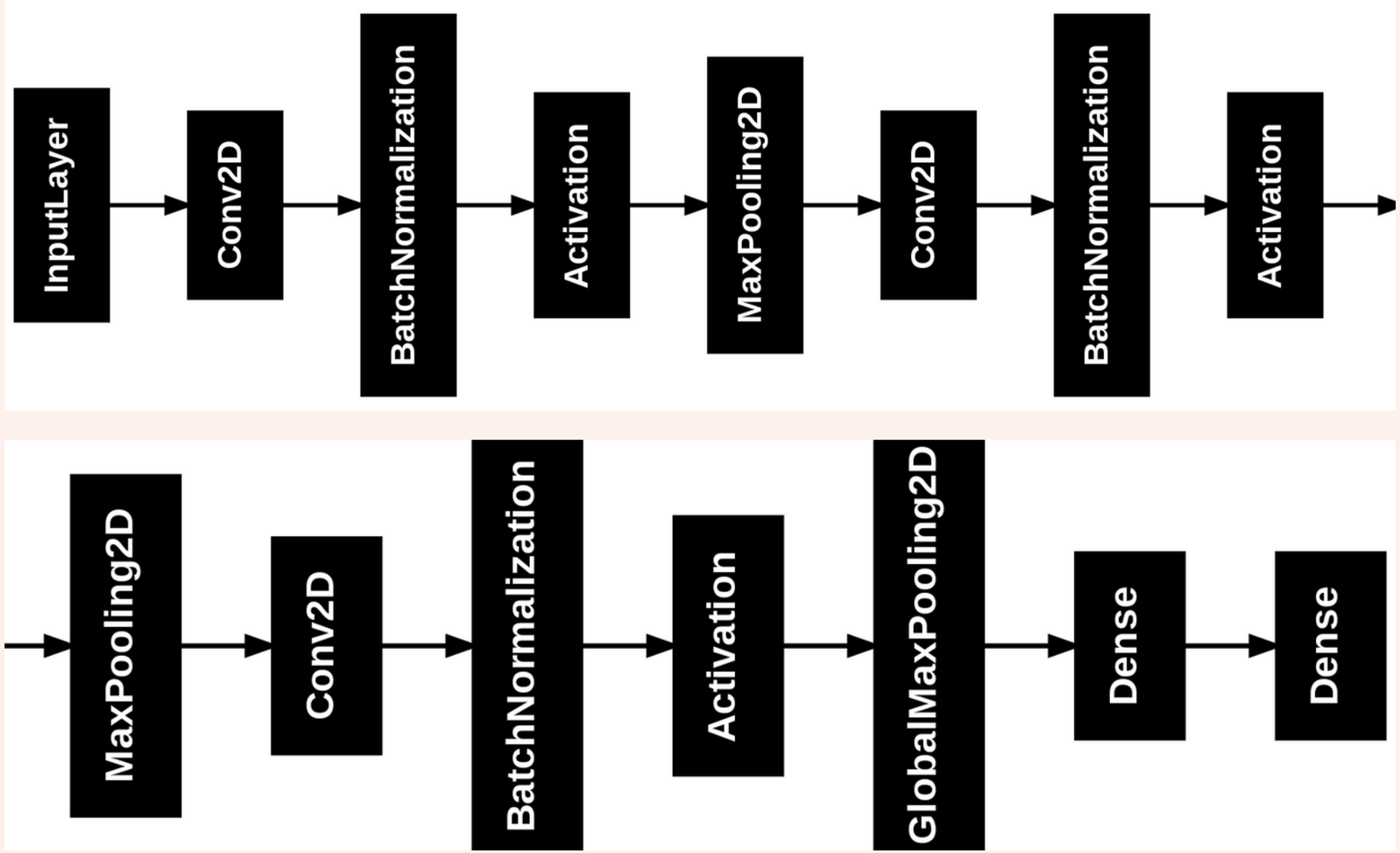
DATA PARTITIONING

The dataset was split into three parts to train and evaluate the model:

1. **Training set**: 70% of the data, used to train the model
2. **Validation set**: 20% of the data, used to monitor performance
3. **Test set**: 10% of the data, used for final evaluation of the model's accuracy



MODEL FROM SCRATCH

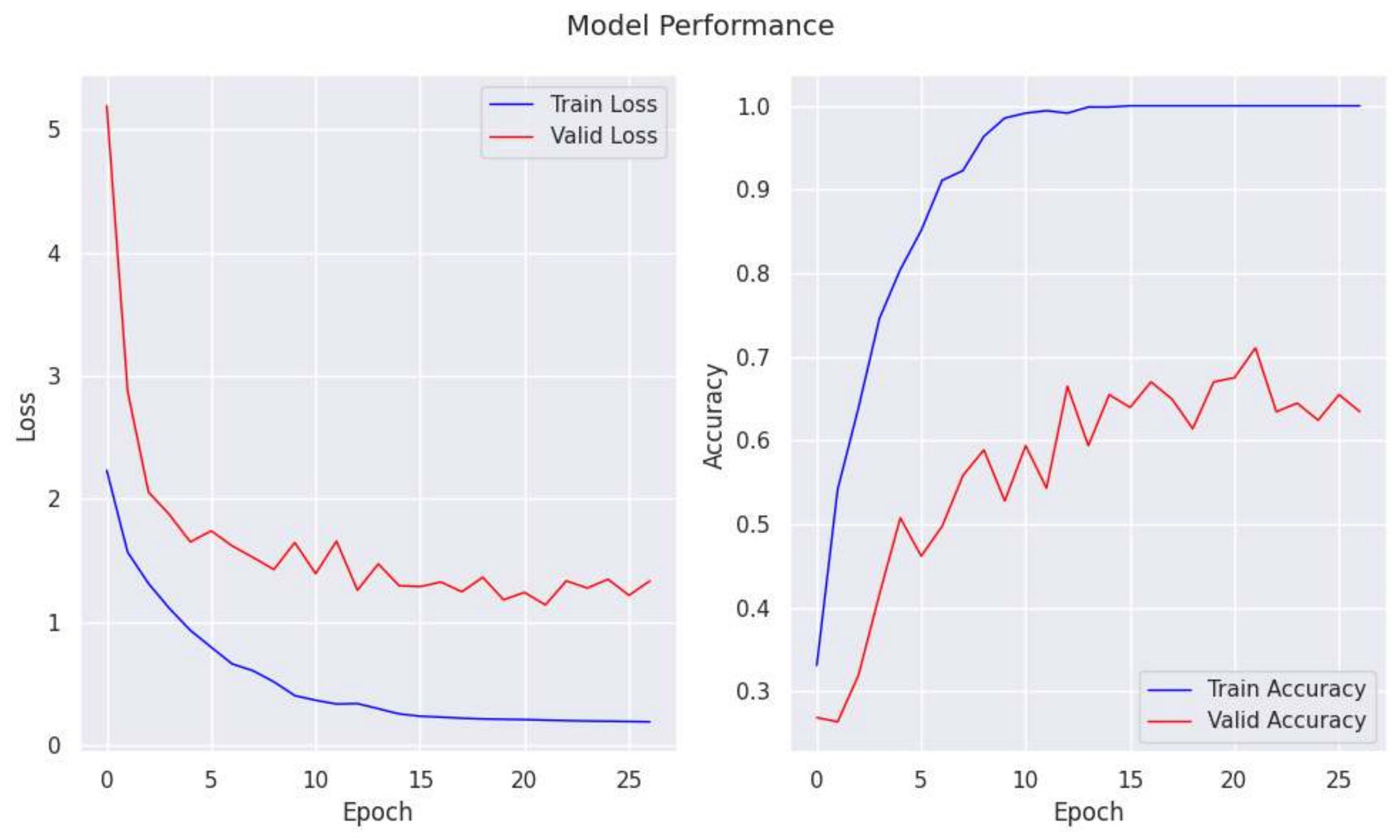


The model is composed of:

- **3 convolutional layers with batch normalization and activation function;**
- **2 fully connected (dense) layers;**
- **kernel regularization.**

Total parameters = 102,985
Trainable params = 102,537
Non-trainable params = 448

MODEL PERFORMANCE



Training

- **Loss:** 0.191
- **Accuracy:** 1.000

Validation

- **Loss:** 1.334
- **Accuracy:** 0.634

Test

- **Loss:** 1.456
- **Accuracy:** 0.615

DATA AUGMENTATION

In order to improve the results a combination of transformation had been applied to the dataset:

- **Random zoom**
- **Random rotation**
- **Horizontal flipping**
- **Translation**

These transformations increase the diversity of the training set without requiring additional data, helping the model learn more robust and features.

MODEL PERFORMANCE + AUGMENTATION

Training

- **Loss:** 0.973
- **Accuracy:** 0.745

Validation

- **Loss:** 1.563
- **Accuracy:** 0.492

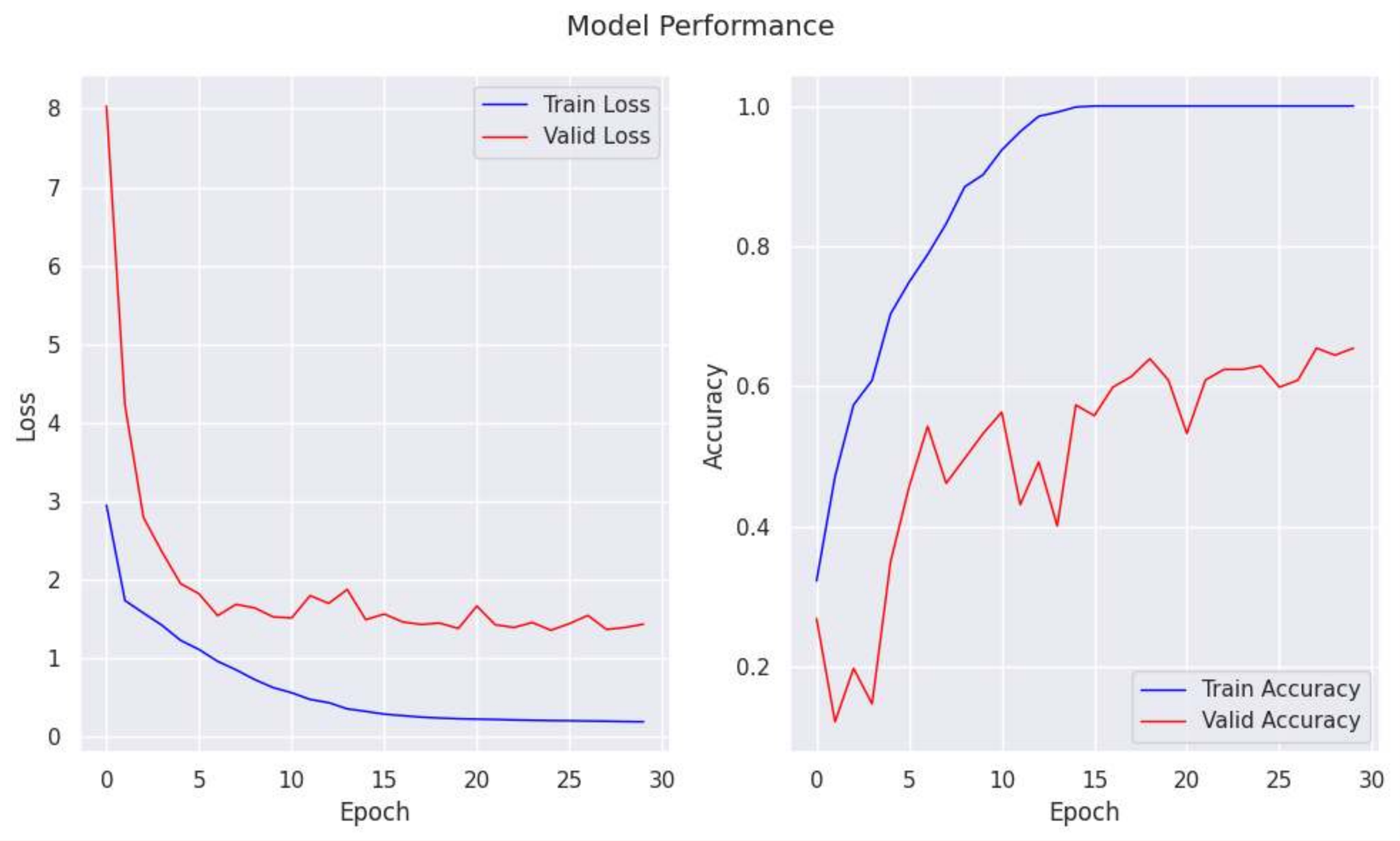
Test

- **Loss:** 1.467
- **Accuracy:** 0.573

DATA SEGMENTATION

Data segmentation technique is a powerful tool for helping the model **learn features** specific to the object.
This should be useful for **reducing** the **overfitting** related to backgroud patterns and **increase robustness** across lighting background or settings.

MODEL PERFORMANCE + SEGMENTATION



Training

- **Loss:** 0.193
- **Accuracy:** 1.000

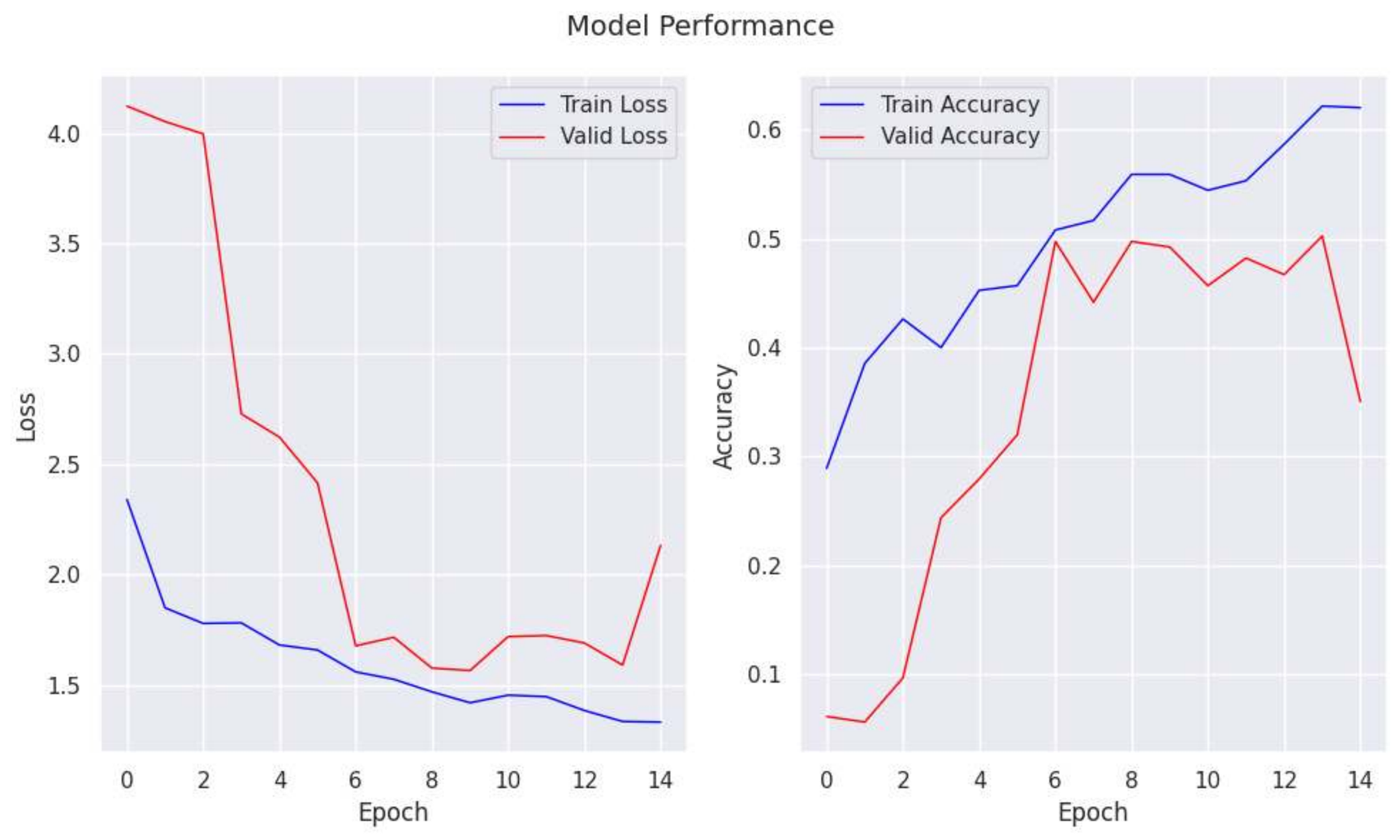
Validation

- **Loss:** 1.436
- **Accuracy:** 0.654

Test

- **Loss:** 2.276
- **Accuracy:** 0.416

MODEL PERFORMANCE + AUG + SEG



Training

- **Loss:** 1.309
- **Accuracy:** 0.633

Validation

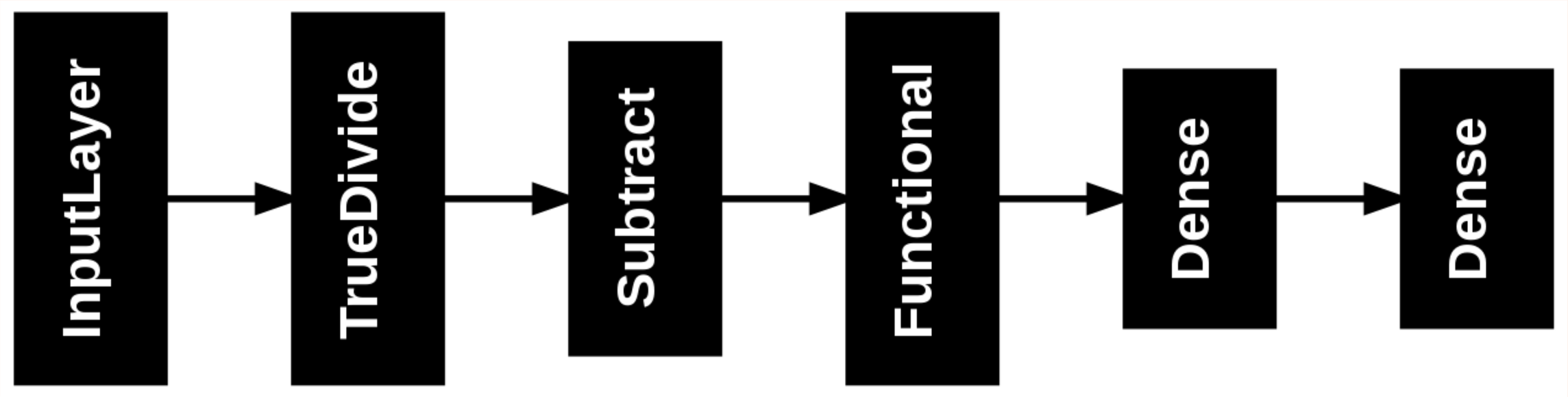
- **Loss:** 2.132
- **Accuracy:** 0.350

Test

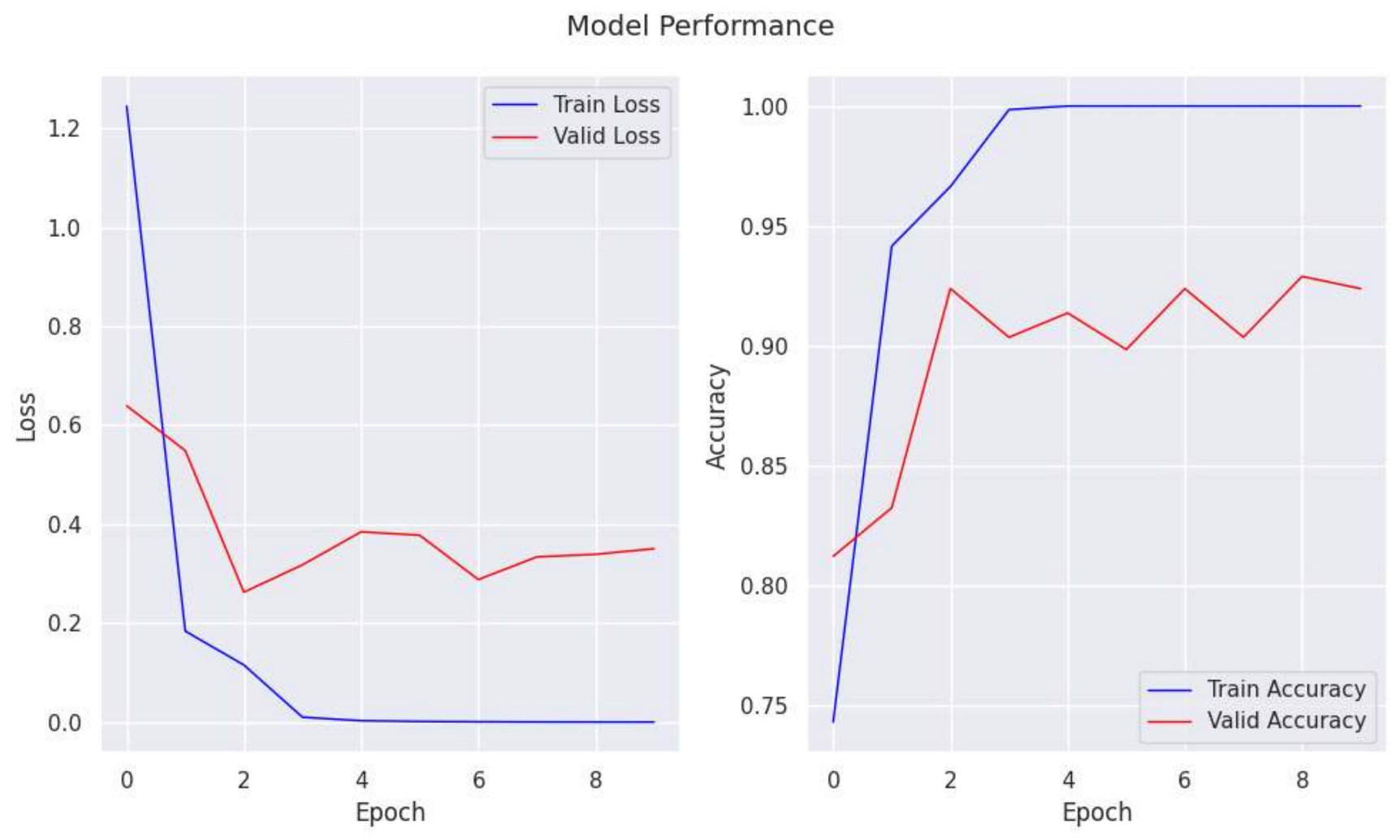
- **Loss:** 0.479
- **Accuracy:** 1.724

TRANSFER LEARNING WITH MOBILENET V2

It has been used the **MobileNetV2** pretrained on ImageNet as a fixed feature extractor, removing its top classification layer and adding custom classifier layers for our 9-category food classification task.



PRE-TRAINED MODEL PERFORMANCE



Training

- **Loss:** 2.744e-04
- **Accuracy:** 1.000

Validation

- **Loss:** 0.351
- **Accuracy:** 0.924

Test

- **Loss:** 0.520
- **Accuracy:** 0.906

CONCLUSIONS

MODEL	TEST LOSS	TEST ACCURACY	FINAL NOTES
Basic CNN	1,456	61,50%	Solid baseline performance from scratch.
CNN + Augmentation	1,467	57,30%	Slight accuracy drop; augmentation may need better tuning.
CNN + Segmentation	2,276	41,70%	Segmentation alone degrades performance, likely due to noise or loss of key features.
CNN + Augmentation + Segmentation	1,724	47,90%	Improved over segmentation alone but still inferior to baseline.
MobileNetV2 Pretrained CNN	0,521	90,60%	Dramatic increase in accuracy; pretrained model benefits from learned features in ImageNet.